

MAHMOUD SALAMA

ENTERPRISE ARCHITECTURE & DIGITAL TRANSFORMATION LEADER

Target-State Architecture | Platform Modernization | Integration & Data Governance | Technology Strategy
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PROFESSIONAL SUMMARY

Enterprise architecture and digital-transformation leader with 18+ years across Egypt and Oman, progressing from hands-on software engineering to ownership of technology strategy, enterprise/solution architecture and national-scale platform transformation. Leads a 54-person technology organization with seven direct reports, translating business capabilities and regulatory constraints into target-state architectures, transition roadmaps, integration models, security/resilience controls and measurable operating outcomes across government and healthcare.

CORE SKILLS

Enterprise & Solution Architecture | Technology Strategy & Roadmaps | Target-State & Transition Architecture | Application Portfolio Modernization | Integration & API Architecture | Data Architecture & Governance | Security & Resilience by Design | Business Capability & Process Architecture | Architecture Risk & Technical Debt | Build-vs-Buy / Vendor Governance | TCO & Investment Decisions | Executive Stakeholder Alignment

SELECTED ARCHITECTURE IMPACT

- Led the ~six-month end-to-end re-architecture and replacement of MediQ - winner of the Pharma Awards 2026 Digital Transformation Award - with a 54-person in-house technology organization.
- Own architecture governance for MediQ, a governmental procurement and supply-chain platform serving **38,000+ users**, **11,000+ healthcare facilities** and **2,000+ suppliers**; an FY2024/25 operational snapshot recorded 89,761 warehouses and 938,242 recipient invoices.
- Architected **MediQ E-Tender** as a governed enterprise workflow spanning **10 major steps**, **30 operational stages**, **15 Chairman approvals** and **20+ roles/permission groups**, with RBAC, segregation of duties, auditability and sensitive-approval controls.
- Directed the completed **EGP 11M Microsoft Dynamics 365** integration across procurement, supply, inventory, finance and reporting, governing data ownership, API/interface contracts, mapping, reconciliation, exception handling, security, change control and production transition.
- Governed the expansion of the MediQ ecosystem into mobile electronic inventory and asset maintenance; designed the geographic asset-mapping architecture using MapTiler and integrated GS1 APIs for GTIN/GLN-based medical-asset identification and location mapping across a maintenance program covering 242,816 medical devices in 164 entities.

PROFESSIONAL EXPERIENCE

Egyptian Authority for Unified Procurement (UPA) | Cairo, Egypt

05/2025 - Present

Head of the Central Administration for Information Systems & Digital Transformation

Scope: Technology Strategy | Enterprise Architecture | Digital Platforms | Integration & Data | Security/Resilience | Delivery Governance

- Own technology strategy and enterprise architecture direction across applications, data, integrations, infrastructure and digital services, with accountability for an **EGP 150M annual IT budget**; define target states, architecture principles, investment priorities, transition sequencing and multi-year roadmaps.
- Scaled the technology organization from **24 to 54 people** and now lead it with **seven direct reports** spanning software/applications, analysis, QA/reporting, DevOps, mobile, data, support/customer service and delivery governance.
- Set and own MediQ's enterprise capability architecture across supply, tendering, inventory, ERP, assets, maintenance, e-pharmacy, complaints and connected health services, aligning target-state boundaries, integration priorities and governance controls.
- Lead architecture and design decisions across engineering, QA, data, DevOps, infrastructure, security and vendors; use PoCs when needed, assess architecture risk, manage technical debt and make build-vs-buy / technology-selection decisions based on TCO, capability and strategic fit.
- Architected and governed cross-government integrations connecting MediQ with ECIC, MoH/HMIS, GAHAC and Blood Bank systems, defining integration design, data flows and technical governance; ECIC and Blood Bank are confirmed production integrations. Separately, personally designed and developed production healthcare REST APIs for secure beneficiary, prescription and dispensing exchange with external health entities.
- Own resilience and production-readiness architecture including capacity planning, performance/load testing, PostgreSQL production/replica architecture, Redis caching, containerized services, **F5 BIG-IP** load-balancing and WAF/edge-security controls, DR environment, **daily full database backups** and off-site recovery, with **manual primary-to-DR failover**.

- Translate legal, regulatory and governance requirements into architecture controls including least privilege, RBAC, segregation of duties, audit trails, traceability, retention, data protection and controlled approvals.
- Own the target 5-layer data/AI architecture spanning operational sources, validation/cleansing, analytical models, executive reporting and phased AI for demand, shortage, expiry and supplier risk.

Egyptian Authority for Unified Procurement (UPA) | Cairo, Egypt
General Manager - Systems, Applications & Technical Support

10/2020 - 05/2025

- Led progressive modernization from established **.NET Framework/Telerik** applications toward **ReactJS/Node.js** services, C#/.NET APIs, **PostgreSQL, Redis, Docker** and **Apache Superset** while preserving operational continuity.
- Converted manual, paper- and spreadsheet-heavy processes into governed digital workflows with role-based access, multi-level approvals, validation, audit trails, reporting and clear operational ownership.
- Owned the full solution lifecycle across requirements and business analysis, solution architecture, engineering, integration/data mapping, QA/UAT, deployment, training, production support and continuous improvement.
- Strengthened SDLC, release and production-readiness controls including architecture/design reviews, engineering standards, test gates, incident escalation, root-cause analysis, documentation and knowledge transfer.

Integral Solutions LLC | Muscat, Oman

05/2016 - 10/2020

Development Team Leader & Senior Full-Stack Developer

- Led **10+ developers** while remaining hands-on across **15+ Omani government platforms**, covering requirements, solution design, development, testing, deployment, releases and production support.
- Contributed extensively to the National Educational Portal serving **approximately 1 million users**, the **Sultan Qaboos Award-winning Recruitment System** and Maktabi electronic correspondence management.
- Designed secure government workflows, reporting, database and integration components across C#/.NET, ASP.NET Web Forms, SQL Server and MySQL.

Keyframe Egypt | Cairo, Egypt

04/2011 - 05/2016

Development Team Leader & Senior Full-Stack Developer

- Delivered or led **150+ major and mid-sized projects** across government, healthcare, education, construction, automotive, events and commercial sectors, excluding small projects; progressed from hands-on engineering to leading a six-person development team.
- Owned analysis, C#/.NET engineering, database/reporting design, testing, deployment and support across multilingual and RTL solutions, including the ITIDA-recognized Yashfeen healthcare solution.

Egyptian Armed Forces | Egypt

04/2008 - 10/2010

Linux Systems & Web Developer

- Built hands-on Linux systems expertise through Linux From Scratch, kernel modules, optimization and hardening, alongside secure ASP.NET/SQL Server application development.

ARCHITECTURE, INTEGRATION & DELIVERY CAPABILITIES

Architecture & Transformation: Enterprise/Solution Architecture, business capability alignment, current/target state, principles/reference patterns, transition roadmaps, portfolio rationalization, technology operating-model evolution, architecture risk and technical debt.

Integration & Data: REST/SOAP, JSON/XML, API contracts, mapping/transformation, internal data models and mapping patterns where applicable, master/reference data, data ownership, quality/reconciliation, PostgreSQL, SQL Server, MySQL, Redis, Power BI and Apache Superset.

Security, Resilience & Operations: Defense in depth, RBAC, least privilege, segregation of duties, TLS, F5 BIG-IP, network segmentation, container security, resilience & DR, backup/recovery, capacity planning, performance testing, monitoring/logging and incident/RCA governance.

Methods & Technology: C#/.NET, ReactJS, Node.js, PostgreSQL, Redis, Docker, REST APIs, OpenAPI/Postman, UML/Visio, Jira, Agile/Waterfall, CI/CD governance and AI-assisted engineering/analysis.

SELECTED ARCHITECTURE & GOVERNANCE CONTRIBUTIONS

- Board Member - National Blood Operations Oversight (2021-2024), representing UPA and contributing to digital governance, data integrity and technology standards.
- Member - Egypt Digital Transformation Committee supporting the World Bank Group B-READY Assessment (2025-2026).

EDUCATION

B.Sc. in Computer Science, Faculty of Computers & Informatics, Zagazig University | 2003-2007 | Arabic: Native | English: Professional Working Proficiency